

LLM sentiment quantification reveals selective alignment with human course-evaluation raters

Joyce W. Lacy^{a,*}, Chi Nnoka^{b,1}, Zachary Jock^c, Cathleen Morreale^d

^a Department of Psychology, University at Buffalo, The State University of New York, Buffalo, 14260, NY, United States

^b Department of Mechanical and Aerospace Engineering, University at Buffalo, The State University of New York, Buffalo, 14260, NY, United States

^c Department of Mathematics, University at Buffalo, The State University of New York, Buffalo, 14260, NY, United States

^d Office of Curriculum, Assessment, and Teaching Transformation, University at Buffalo, The State University of New York, Buffalo, 14260, NY, United States

ARTICLE INFO

Keywords:

sentiment
Human-AI alignment
LLMs
Course evaluation
qualitative data

ABSTRACT

Student course evaluations contain rich qualitative feedback in the form of comments written in response to open-ended questions. However, this qualitative data, which may be more nuanced and detailed than quantitative ratings, is often unexamined in both administrative and research settings due to the labor-intensive nature of manual analysis. We investigate whether large language models (LLMs), including BERT, RoBERTa, and OpenAI model variants, can accurately replicate human judgments of sentiment in these comments. We compare masked and generative language models, using both naive and fine-tuned approaches, to analyze a curated dataset of 1000 de-identified course evaluation responses. Results show that some artificial intelligence (AI) models can approach inter-rater reliability with humans remarkably well and quickly with limited tuning or training data provided. However, performance varied and not all models were able to produce a reliable sentiment analysis, even after training. This has implications for future avenues of qualitative data analysis within course evaluations as well as the large repositories of course evaluations available at institutions of higher education. Importantly, consideration should be taken when selecting an AI model as this decision has ramifications for the reliability and validity of the generated output.

1. Introduction

1.1. The role of course evaluations in higher education

Course evaluations, in which university students rate their perceptions of a course and its instructor, were developed in the 1920s (Stroebe, 2020; Wang & Williamson, 2022). Because they were originally created with the intention of guiding improvements in teaching – if such improvements were needed – access to the evaluations was initially limited to course instructors. However, academic administrations have increasingly seen these evaluations as sources of information from which to gauge tenure worthiness and quantify an instructor's teaching effectiveness, often by comparing them to other instructors. In a survey of college deans in 2000 ($n = 506$) and 2010 ($n = 401$), there was nearly unanimous agreement (97.5 % and 99.3 % of participants, respectively) that classroom teaching was a “major factor” in evaluating overall faculty performance, and student evaluations were the top source of

information (88.1 % and 94.2 %, respectively) used in evaluating teaching (Miller & Seldin, 2014).

Whether or not course evaluations should be used in this manner is debated as some have postulated that student ratings are more accurately a measure of “student satisfaction” than they are of “instructor effectiveness” (Wang & Williamson, 2022). Several studies have noted the correlation between student grades and instructor evaluations (e.g., Cho et al., 2015) and, given the trends in both over the past half-century, have theorized that course evaluations contribute to grade inflation (Stroebe, 2020; Wang & Williamson, 2022). In one study, roughly a third of student survey responders agreed that some students give instructors low ratings as punishment for low grades (Backer, 2012).

When formally surveyed, instructors typically report positive but cynical views about the utility of course evaluations (Beran & Rokosh, 2009). Most respondents view them as useful for crafting lectures and refining course materials, as well as general feedback on the overall quality of their teaching, but do not find them helpful for selecting

* Corresponding author.

E-mail addresses: jlacy@buffalo.edu (J.W. Lacy), chinnoaka@buffalo.edu (C. Nnoka), zjock@buffalo.edu (Z. Jock), cwalther@buffalo.edu (C. Morreale).

¹ These authors contributed equally to this work.

textbooks or creating exams or assignments. Many instructors are skeptical of their use in retention, promotion, and salary decisions, and not all agree that administrative heads should have access to course evaluation data (Beran & Rokosh, 2009). Interestingly, students themselves do not necessarily enjoy the course evaluation process (Backer, 2012).

Despite the lack of consensus on the utility of course evaluations, it is undeniable that they have become commonplace in university settings. In 1973, of the colleges surveyed, 29 % collected course evaluations. That number increased to 68 % in 1983, 86 % in 1993, and 94 % in 2010 (Miller & Seldin, 2014; Seldin, 1998). This ubiquity has naturally led to an exploration of factors that influence these evaluations. These factors typically fall into one of three categories: instructor, course, or student variables. Regarding instructor factors, most studies have examined either gender or minority status. There is some evidence that female instructors receive lower course evaluations than their male counterparts (Mengel et al., 2019). However, this has not been a consistent finding. Many studies have found either no difference or negligible differences with trivial effect sizes (Centra, 2009; Li & Benton, 2017). Some researchers have proposed that this pattern of results may be due to an interaction between gender, discipline, and course level. In disciplines with large gender disparities, students may develop expectations of who is qualified to teach lower-versus upper-level courses. Instructors who violate these expectations typically receive lower evaluations (Aragon et al., 2023). Compared to the effect of gender, there is slightly stronger evidence that non-White instructors receive lower evaluations than their White peers (Smith, 2007; Stroebe, 2020).

Unlike instructor variables, there are more consistent findings regarding course factors. For example, multiple studies have shown that non-STEM courses are rated more favorably than STEM courses (Brown et al., 2020; Centra, 2009; Hoyt & Lee, 2003; Li & Benton, 2017; Wang & Williamson, 2022). Some research studies have also looked at class size. There is a modest negative correlation between class size and both course and instructor evaluations, such that smaller classes and instructors of smaller classes are rated more favorably (Benton & Li, 2015; Centra, 2009; Hoyt & Lee, 2003). It should be noted, however, that in these studies, a “small” class was 10–14 students. A class of 15–34 students was considered “medium”, 35–49 students was considered “large”, and 50 or more students was considered “very large”. Thus, these findings may not generalize to large universities where it is not uncommon to have classes of 100 or more students.

Student variables are difficult to study in course evaluation research due to the confidential nature of evaluations. Investigators rarely know a student's gender, race, or even college year. However, some studies have investigated student factors self-reported on the course evaluations. For example, students who are motivated to take a course and students who self-report above average study habits tend to rate courses and instructors more positively (Benton & Li, 2015; Hoyt & Lee, 2003; Li & Benton, 2017). Moreover, students report a greater desire to take a course when it is part of their major as opposed to a general education requirement (Benton et al., 2015; Benton & Li, 2015), though there are mixed results regarding whether a student's prior interest in a topic is correlated with more positive evaluations (Stroebe, 2020).

1.2. The challenge of analyzing qualitative feedback

The great majority of research conducted on course evaluations has focused on quantitative data (e.g., ratings on 5-point Likert-scales; Likert, 1932) while the qualitative student comments written on evaluations has largely been overlooked (Steyn et al., 2019). There are inherent differences in quantitative and qualitative data. For example, while it is easier to conduct statistical analyses on quantitative data, it is typically less detailed and nuanced than qualitative data. Quantitative data can be biased by a response set, which is the tendency for a participant to answer all questions the same way (e.g., all “strongly agree” or all “strongly disagree”), or range effects, such as ceiling or floor effects,

when all of the responses are clustered near one end of the range of possible responses (Patten, 1998). Quantitative data, which is constrained by predetermined response options, may be more artificial and nonrepresentative than qualitative data where participants are allowed to respond in their own words.

When it comes to course evaluations as a metric of an instructor's teaching effectiveness, the limitations of quantitative data add to the previously mentioned list of concerns regarding the validity of student ratings. Analyzing the qualitative data (i.e., written student comments) of course evaluations might address both sets of concerns. Qualitative data is not affected by response sets or range effects, and since qualitative responses are typically more detailed and nuanced, they may be more informative than quantitative ratings. As an illustration, Student A who writes, “This instructor was horrible! I've never gotten a such a low grade in a class before!” and Student B who writes, “This instructor was horrible! His lectures were so disorganized, and he was condensing to students who asked questions in class!” may give their instructors equally low ratings (e.g., a 1 out of 5), but their rationale for doing so differs. The first case seems to be more indicative of grade “satisfaction” while the second case may be a more valid assessment of instructor “effectiveness”. Of course, qualitative data is not without limitations. Student C could simply write, “This instructor is horrible!” without any additional context. The ambiguity of qualitative responses is one of the challenges of qualitative data analysis.

The biggest obstacle to qualitative data analysis is that it is time consuming and labor-intensive (Pfeifer & Dolan, 2023). Traditional analysis of qualitative data typically involves identifying themes, defining these themes, and training several research assistants to code the data by these themes (Flick, 2009; King & Brooks, 2018; Silverman, 2000). Pfeifer and Dolan (2023) estimate 400 coding hours for a research study with 30 participant interviews. This type of manual analysis has not been attempted in course evaluation research. Thus, it is difficult to estimate the time needed. However, course evaluation research on various instructor, course, and student factors typically has large datasets encompassing multiple courses and, sometimes, across multiple semesters. Thus, the time and labor required becomes a significant obstacle, especially at institutions and in disciplines that have large enrollment classes.

Artificial intelligence (AI) models offer a potential solution to this problem. Recent studies have demonstrated that AI can be trained to perform as well as human coders of qualitative data (Hamilton et al., 2023), in some cases outperforming humans (Tornberg, 2023). In the context of course evaluations, which are inherently a subjective, human-based opinion, the AI model will need to perform a sentiment analysis of the student comments written on course evaluations (Onan, 2020). If AI models are able to assess sentiment as accurately as human coders, this opens many new potential avenues of research, especially since an AI can process tens of thousands of student comments without the limitations of fatigue.

1.3. Natural language processing and large language models

Natural language processing (NLP) is a branch of artificial intelligence that equips computers to interpret and generate human language tasks. NLP is usually grouped under natural language understanding (NLU), the extraction of meaning from raw text, and natural language generation (NLG), the production of coherent text given some representation or intent (Kamath et al., 2024).

At its core, a language model assigns probabilities to how a text will continue: given a string of words (e.g., the token prefix $w_1, \dots, w_n$), it estimates the likelihood of each possible next word (i.e., token). Classic n -gram models implement this idea by computing conditional word probabilities from corpus counts, ranking the most likely continuations (Jurafsky & Martin, 2008; Shannon, 1948). A pivotal advancement in this field came when Bahdanau et al. (2016) introduced an attention mechanism that lets a decoder weigh different parts of its input

dynamically. Vaswani et al. (2017) elevated this idea into the *transformer* architecture, replacing recurrence entirely with multi-head self-attention and feed-forward blocks. While older models processed text linearly (one word at a time), the "attention mechanism" allows the model to simultaneously weigh the importance of all words in a sentence regardless of their distance from one another. This enables the model to resolve ambiguities, for example, by "attending" to the word "film" to understand that the word "great" refers to cinema quality rather than size. By focusing selectively on the most relevant tokens, transformers accelerated training and delivered dramatic quality gains across translation, summarization, and many other tasks (Vaswani et al., 2017).

Transformers can process vast bodies of text in parallel, inspiring a two-stage workflow: first *pre-train* a general-purpose language model on raw text, and then *fine-tune* it for a downstream task. This recipe is now standard across NLP. As parameter counts and data sizes rose from millions to hundreds of billions, the same transformer core began to exhibit qualitatively new behaviors such as few-shot in-context learning, code generation, and analytic reasoning (Brown et al., 2020). Models at this scale are collectively termed *large language models* (LLMs). Current LLMs fall into two broad families. Masked language models (MLMs) are trained to predict randomly masked tokens (essentially "filling in the blanks" within a sentence), yielding strong fixed-length representations that excel at understanding the context of a whole passage for classification. Generative, or causal, language models (GLMs) predict the next token in a sequence to produce new text, enabling zero-shot and few-shot prompting where the model performs a task based on a simple instruction without prior specific training. In the current study, we compared four AI models: two are MLMs, BERT and RoBERTa, and two are GLMs, OpenAI's GPT-4o and o3.

1.4. Masked language models: BERT and RoBERTa

Google's BERT (Bidirectional Encoder Representations from Transformers), released in 2018, introduced the masked-token and next-sentence-prediction (NSP) objectives and was pre-trained for four days on BookCorpus (800 million words) plus English Wikipedia (2.5 billion words), yielding 110–340 million weighted parameters that could be fine-tuned (i.e., adjusted) with only a few task-specific steps (Devlin et al., 2019). In contrast to unidirectional models (i.e., language models that only read a sentence in one direction), BERT was developed with the capability of understanding the context of words by examining both sides of a given text sample at once. The model's bidirectional representations set new state-of-the-art results across NLP benchmarks like GLUE, SQuAD and SWAG, making it a staple for sentence-level sentiment, named entity recognition (NER), and question answering tasks (Devlin et al., 2019).

To achieve the ability to understand language context, the pre-training process for BERT considers two objectives: the random masking of words from a given text sample (the MLM objective) and the ability for the model to predict two sentences following each other in the original text (the NSP objective). The MLM objective trains BERT to guess the appropriate word that fits in each sentence during a training sequence (Liu et al., 2019). In the context of sentiment analysis, BERT has each word categorized based on whether it is negative, neutral, or positive (accounting for the broader context of the sentence). The scores for each word are summated into a single numerical score for the entire sentence (Kebede & Tesfai, 2023).

RoBERTa (A Robustly Optimized BERT Pretraining Approach) is an alternate version of BERT developed by Meta in mid-2019 (Liu et al., 2019) with the intention of streamlining BERT's language processing features in pursuit of better performance in downstream tasks (e.g., sentiment analysis). RoBERTa kept BERT's encoder-only architecture but removed next-sentence-prediction, as it was determined that removing the NSP objective led to the same if not slightly better task performance. Unlike BERT, the MLM objective of masking a random word in a sentence is done on a per-sequence basis rather than at the

start of the training instance, which allows for stronger inference in language context needed for large datasets. The dataset used to pre-train RoBERTa was substantially larger than BERT with ~ 160 GB of text drawn from Wikipedia, BookCorpus, CC-News, OpenWebText and Stories (Gokaslan & Cohen, 2019; Nagel, 2016; Trinh & Le, 2019; Zhu et al., 2015). This extended amount of pre-training combined with higher batch sizes during training sessions improved task accuracy, in part to the redesigned MLM objective, and delivered 4–5 percentage-point average gains on the GLUE benchmark at the cost of longer training and greater energy use (Liu et al., 2019). Both MLMs excel at fixed-length classification, but their 512-token context window and lack of native generation (i.e., the inability to generate arbitrary text) limit applicability to open-ended dialogue or summarization tasks.

1.5. Generative language models: GPT-4o and o3

OpenAI's Generative Pre-trained Transformer 4 Omni (GPT-4o) was announced May 2024 as the first flagship model trained jointly on text, images, and audio (Hurst et al., 2024) and is the current default LLM used by the chatbot ChatGPT. The unified multimodal stack enables real-time speech-in/speech-out interaction with roughly half the latency and cost of GPT-4-Turbo while matching its text benchmarks (Hurst et al., 2024; Seetharaman, 2024). Released in December 2024 and expanded in April 2025, the text-only o3 family reallocates parameters from the multimodal pathways to deeper chain-of-thought reasoning layers (OpenAI, 2025). Benchmarks show o3 surpassing GPT-4o on math, code, and analytic tasks while remaining cheaper per token and offering adjustable "reasoning effort" settings (Kerner, 2025). In practice, GPT-4o is preferred when screenshots, diagrams, or voice memos accompany input, whereas o3 is the economical choice for large-scale, purely textual sentiment sweeps. The reported trade-off is GPT-4o's multimodal reach versus o3's leaner footprint and stronger step-by-step reasoning. However, compared to BERT and RoBERTa, which are open source, GPT-4o and o3's closed code results in greater opacity regarding how the LLM is responding to task instructions.

1.6. Research questions of the current study

The challenges of analyzing the qualitative student comments in course evaluations and the distinctions between different AI models motivate the following research questions that guide our study.

RQ1: Can AI generate sentiment ratings of student comments written on course evaluations equivalent to that of human raters?

RQ2: Will the AI-generated ratings improve (i.e., be more consistent with human raters) if the models are trained, or fine-tuned, specifically on course evaluation data?

RQ3: Are there differences in performance in sentiment analysis across different types of AI models?

With the many different modalities of AI that exist, and a fast-growing incorporation of AI into higher education in several facets (Crompton & Burke, 2023), it is all the more paramount to conduct detailed investigations into the viability of these AI models for applications like sentiment evaluation, a staple of higher education. Hence, the motivation for our research study.

2. Methods

2.1. Overview

Before we can address RQ1 (whether or not AI can generate sentiment scores comparable to human raters), we first need to curate a representative set of course evaluation comments and acquire human sentiment ratings of those comments. We also need these human sentiment scores to train and fine-tune the AI models for RQ2. Lastly, we

perform these steps for multiple models in order to answer RQ3. Thus, our entire study (Fig. 1) encompasses four stages: 1) curating a list of real student comments from past course evaluations, 2) acquiring human sentiment ratings of these comments, 3) generating AI sentiment ratings of these comments, and 4) comparing the human ratings to the AI ratings.

2.2. Curating a representative set of course evaluation comments

To curate a set of course evaluation comments, we retrieved existing course evaluation data from an institution's Spring 2018 semester. Students were prompted to provide written responses to a few different questions on their end-of-the-semester course evaluations. However, we focused on the “overall instructor effectiveness” question (i.e., “Please comment on how effective the instructor was in teaching this course”) since “instructor effectiveness” seem to be the primary focus of higher education institutions when making decisions regarding the hiring and promoting of faculty (e.g., Wang & Williamson, 2022).

We narrowed down the list of courses that student comments came from to 16 courses spread over 14 departments. Our decision criteria for choosing courses included cumulatively representing a wide array of disciplines, including 100-level to 400-level classes to encapsulate the entire college experience, and lastly, selecting classes with relatively large enrollments so that there would be enough course evaluation comments to work with. Some examples of the courses these comments came from include BIO 318 Genetics, CHE 204 Organic Chemistry, CSE 421 Operating Systems, MTH 241 College Calculus III, ENG 100 Writing and Rhetoric, HIS 317 History of Early Modern Britain, PHI 115 Critical Thinking, PSC 303 Constitutional Law, and SOC 349 Classical Sociological Theory.

We aimed to curate a final list of 1000 individual student comments for the purposes of this study. Thus, the last stage of selection involved removing extra copies of commonly repeated comments (e.g., “very good”) and pseudorandomly selecting comments from classes so that there was a relative similar number of comments taken from each course in the overall dataset. Half of the set (i.e., 500) came from STEM courses and half from non-STEM courses. Lastly, we deidentified the final set of 1000 comments and removed any personal information (e.g., instructors' names were removed). Otherwise, the student comments were left unedited. Typos and spellings errors were not altered.

2.3. Acquiring human ratings

Whether a comment is “good” or “bad” is inherently a subjective and human-based decision. Thus, we needed human sentiment ratings of these comments to compare our AI ratings to, as well as to train and fine-tune the AI models. To this end, we created a Qualtrics survey that

presented the course evaluation comments described in the previous section to current undergraduate students. Prior to any data collection, this study was approved by the institutional ethics committee (IRB ID #00008366). Informed consent was obtained from all participants, and their privacy rights were strictly observed.

2.3.1. Survey design

Participants were instructed that they would view a series of real student comments written in response to the “instructor effectiveness” question of a course evaluation and that their task was to rate whether they considered the comment to be positive or negative on a 7-point scale ranging from -3 to $+3$. They were also informed that instructor names were removed but that, otherwise, the original student comments were unedited and, therefore, may include misspellings and typos. Participants were not given any other information about the courses or instructors that these comments were in reference to.

A natural limitation of human raters of qualitative data is fatigue. As humans process more and more information, boredom or monotony may degrade the quality of their decision-making. To address this issue, we estimated that an individual participant would only be able to rate approximately 100 comments, but that collectively across many participants, we could quantify an average human rating for each of the 1000 course evaluation comments.

The order of the curated list of 1000 comments was randomized and then broken into 10 groups of 100 comments each. A participant only viewed the comments from one group, and the order of these comments was also randomized on an individual basis. These layers of randomization aimed to prevent a participant's ratings from being biased by exposure to a specific subset of classes (e.g., only being given comments from biology and chemistry classes) or by exposure to a certain order of comments (e.g., rating 50 comments from STEM courses before rating 50 comments from non-STEM courses).

2.3.2. Participant recruitment and demographics

Participant recruitment and data collection occurred during the Spring 2025 semester. Participants were selected from the enrollment lists of 14 different undergraduate courses, which represented the same 14 disciplines encompassed in the curated set of 1000 course evaluation comments. Effort was made to select the same courses from which the comments were taken from (e.g., CHE 204 Organic Chemistry, MTH 241 College Calculus III, etc.), but in some cases, a different course within the discipline was used to recruit participants (e.g., BIO 201 Cell Biology, HIS 209 The American Civil War, etc.).

A pseudorandom set of potential participants from these class enrollment lists were contacted by the principal investigator via email with an invitation to participate in the Qualtrics survey. Effort was made to invite a relatively similar number of participants from each class.

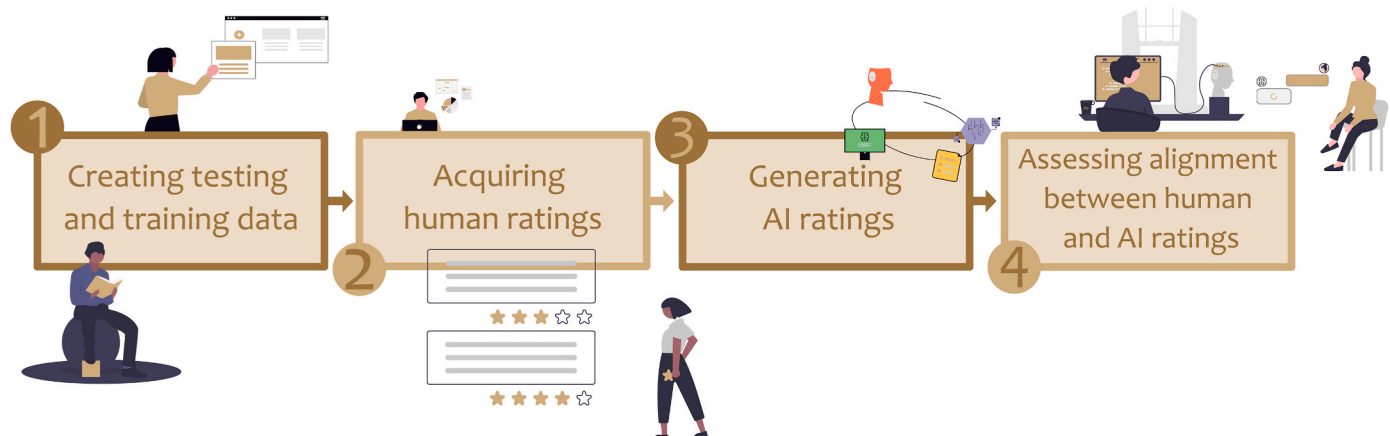


Fig. 1. An overview of the entire study.

Thus, in the largest of these classes, not all enrolled students were invited to participate. No other criteria were used to select participants besides enrollment in these courses.

A total of 172 participants completed the survey in its entirety. Three manipulation check questions were built into the survey to capture inattention to the survey questions. Eighteen participants failed two or more of the manipulation checks and were removed from the final data set, leaving a total of 154 participants (103 females, 46 males, 5 participants did not identify as either male or female). The median survey completion time was 20.27 min. All participants who completed the survey were compensated with a \$5 gift card to [Amazon.com](https://www.amazon.com). The average age of participants was 19.69 ($SD_{age} = 1.35$); 14.9 % of the participants were freshmen, 26.6 % were sophomores, 29.2 % were juniors, and 29.2 % were seniors, as defined by the number of college credits completed. Participants were allowed to identify as multiple races or ethnicities: 18.2 % of participants selected more than one race or ethnicity. In total, 50.7 % of participants identified as White, 14.9 % as Black or African American, 9.1 % as Hispanic, 25.3 % as East Asian, 14.3 % as South Asian, 3.9 % as Middle Eastern, 1.3 % as Indigenous American, and 0.7 % as Pacific Islander.

Because each participant only rated one tenth of the entire curated set of course evaluation comments, this means each comment was rated by an average of 15.4 participants (range 13–19). Their scores were averaged per comment to create a list of human sentiment ratings for each course evaluation comment. This would become the data for which we tested and trained our AI models.

2.4. Generating AI ratings

Our AI sentiment analysis investigated two families of LLMs: MLMs and GLMs. MLMs were used because of their ability to improve understanding and generation of human language via their prediction of missing words in a sentence through analyzing context of surrounding words and phrases. GLMs were investigated because of their abilities to understand language patterns and learn from vast amounts of text data rapidly, a feature that could be useful when considering AI for use in sentiment analysis of thousands of qualitative student responses from a myriad of classes of any given institution. The MLMs used in this paper were Bidirectional Encoder Representations from Transformers (BERT) and Robustly Optimized BERT Pretraining Approach (RoBERTa), whereas the GLMs used herein were the Generative Pre-trained Transformer 4 Omni (GPT-4o) and o3 from developer OpenAI.

Comparing these MLMs and GLMs also allowed us to ascertain the difference in performance from AI techniques with varying levels of automation and transparency. Generative language models like OpenAI's GPT-4o and o3 are very user-friendly and have a high degree of automation. One does not need to know how to code in computer languages and can simply type requests into a prompt window, greatly reducing the amount of hands-on work required by a human operator. Additionally, ChatGPT (OpenAI's GPT-4o) is currently one of the best known and widely used LLM chatbots amongst the general public, and thus, assessing its performance is integral to the themes of our research study. However, much of the backend of how these GLMs achieve their output, or how exactly they were initially trained is either not easily interpretable or is actively hidden behind the walls of parent companies that may be incentivized to be closed source in order to compete with other generative AI companies, further degrading transparency and making technical progress more challenging (Liu et al., 2023). Masked language models like BERT and RoBERTa represent much more automation than traditional manual sentiment scoring done by humans, but less automation than that of generative language models. There is still a lot of user input behind MLMs such as the selection of transformers used (which have publicly available details on how they were initially trained) or finely tuning the quantity and quality of subsequent training based on custom input datasets. However, advantages of these MLMs include their customizability and a much higher level of transparency

than OpenAI's GPT-4o and o3; BERT and RoBERTa are both open source and thoroughly documented (e.g., Devlin et al., 2019; Kebede & Tesfai, 2023; Liu et al., 2019).

2.4.1. BERT and RoBERTa's sentiment analysis

To increase the accuracy of BERT and RoBERTa's sentiment analysis capabilities for this study, we had both models undergo the process of fine-tuning, wherein text samples are given to the model as training input, and their weighted parameters are adjusted according to the new text and accompanied sentiment scores. The fine-tuning process allows for a curated version of BERT and RoBERTa that is receptive to the language nuances of the dataset and can be subsequently fine-tuned with additional text samples at a later time to reflect evolving language. By default, BERT and RoBERTa's classification component does not reflect the language nuances of our dataset (i.e., student comments on course evaluations) despite their extensive pre-training, so we will consider them "naïve". The sentiment analysis output for both naïve BERT and RoBERTa are in juxtaposition to the sentiment analysis output for the fine-tuned versions of the models to determine if fine-tuning provides better correlations with human ratings, which is our RQ2.

The curated list of 1000 course evaluation comments was split into two separate files. The first file, containing the 500 even-numbered comments, was used to benchmark BERT and RoBERTa's performance across naïve and fine-tuned versions. We refer to this as the testing set of comments as the AI models' sentiment scores on these comments would be compared to human ratings of the same comments. The second file, which was our training set of comments containing the 500 odd-numbered comments, was used to fine-tune 10 separate cases of both BERT and RoBERTa by comparing its sentiment score output to a specified sample size of responses and human-rated sentiment scores. The 10 separate cases represented different amounts of training which varied from 50 comments to 500 comments. The order of the list of comments was randomized, thus there should be no systematic differences between even- and odd-numbered comments. Even and odd enumeration was simply used as a quick rule to divide the list into testing and training data.

BERT and RoBERTa were downloaded from HuggingFace (HF), which is an online community that is dedicated to hosting AI models and training datasets created by users from around the world. In addition to its hosting capabilities, HuggingFace's API (Application Programming Interface) allows users to download models and/or datasets from the website using the Python programming language to communicate with HF servers. To import a model from HF, the user can search the name of the model on the website and copy the name of the model into a Python script or command. For BERT, *google-bert/bert-base-uncased* was used (*bert-base-uncased* has all of its pre-training stripped of capital letters to remove bias for proper nouns). For RoBERTa, *unso/roberta-large-fine-tuned-ss5* was used. This RoBERTa model was fine-tuned using the Stanford Sentiment Treebank with 5 labels (SST-5) dataset containing over 6000 movie reviews (Tai et al., 2015).

For the fine-tuning process, BERT and RoBERTa were both trained using the training set of course evaluation comments accompanied by human sentiment rating scores. As mentioned, we varied the amount of training the AI models received in 10 separate cases, from 50 to 500 comments. For each case being trained, a portion of the responses and sentiment scores would be selected as a basis for BERT or RoBERTa to adjust its weighted parameters, with the rest of the responses being used to evaluate the performance of the fine-tuned model in 10 iterations (or "epochs") before ultimately saving the best version of the new model as a local file directory to be used for batch sentiment scoring of the testing comments at a later time.

To output a sentiment score for a course evaluation response, the text sequence was sent to a Python script that factored the sequence into its individual words, where they were given a probability of belonging to 1 of 5 sentiment bins (very negative, negative, neutral, positive, very positive) before being summated into a single sentiment score for the

entire comment. A 5-point scale was chosen for our AI models, as opposed to the 7-point scale used in the human ratings, because SST-5, which RoBERTa was pretrained on, only includes a maximum of five labels. The 7-point scale from the human sentiment scores was translated into a 5-point scale to provide appropriate feedback to the AI models when fine-tuning. To batch process the course evaluation responses, a second Python script that processes each comment by row and attaches a sentiment score in an adjacent column was used.

2.4.2. GPT-4o and o3's sentiment analysis

The sentiment analysis of GPT-4o and o3 was conducted in a similar manner to the analysis of BERT and RoBERTa. Sentiment scores on a 5-point scale were first acquired from each naïve model for the testing set of course evaluation comments. Then 10 separate cases of fine-tuning occurred, each case representing a different amount of training varying from 50 to 500 comments and their human-rated sentiment scores. The major difference between the sentiment analyses of the GLMs and MLMs is that, instead of using a Python script, instructions were fed directly to the chatbot. For the naïve study using GPT-4o, the following prompt was fed into the chat window with no other commands:

Please read the following.csv file with 500 comments and rate the sentiment of each comment on a 5-pt Likert scale of 0–4, with 0 being the most negative and 4 being the most positive. Append a new column to the testing set's.csv file, and include in it your ratings for the corresponding comments, and please output the new.csv file after you complete your evaluation. Thank you.

This prompt was used with the o3 model as well, and then similarly used for the fine-tuning analyses with some slight tweaks. In cases of fine-tuning, two.csv files were fed to the chatbot with instructions to use supervised learning on the training comments to rate the testing set. Each case that was run began in new chats with memory turned off so that each case of various amounts of training data was independent of each other case and not influenced by previous prompts in the chat window.

2.5. Assessing sentiment quantification alignment between humans and AI

Once the human ratings and the AI ratings were both gathered, a systematic effort to compare the alignment of these sentiment quantification endeavors was needed. For that methodological purpose, we implemented Pearson's correlation coefficient (*r*), a statistical measure that evaluates the strength and direction of linear correlations between two sets of data. While the mathematical foundations of the correlation surface were established by Bravais (1846) and the conceptual framework of "co-relation" was pioneered by Galton (1888), the product-moment formula utilized in this study follows the formalization by Pearson (1896). The resulting *r* values provide the statistical basis for the following analysis, which categorizes the specific instances where LLM sentiment quantification demonstrates robust alignment versus areas of significant divergence. Moreover, the Pearson correlation implementation here at this scale offers quite useful implications for an educational/pedagogical framework given the fact that these models possess the ability to handle high volumes of assessment data frequently found in higher education. The results of our alignment comparisons are detailed in the subsequent section.

3. Results

3.1. Human ratings

The average participant rating for each comment spanned the range of possible scores [−3 to +3]. However, there was a positive skew. The mean average rating was 0.979 (*SD* = 1.47) and 70 % of comments fell within the top two most positive tiers, when the entire range is divided into five categories (see Table 1 for examples of comments). The positive skew may reflect a bias in the raters (e.g., viewing each comment in a

Table 1
Examples of course evaluation comments.

Range of ratings	Examples of comments that fell into each category
Most negative ratings [−3 to −1.8)	"Absolutely awful. I would like to say I did well in this class before you think that I was just another person who did poorly and is saying this stuff out of spite. I have never dealt with anyone with such an ego teaching stuff that they invented as well as saying that people from other established institutions are wrong and then offering his opinion as if it was the holy grail of answers. All the while offering no legitimate arguments other than "I am right" or "I can't hear you""
65 comments	"Spoke very monotoned as if he didn't want to be teaching the course"
Second most negative ratings [−1.8 to −0.6)	"Had a lot of errors and unorganization during class. The clickers mentioned at the beginning of the year never happened. Homework question keys had multiple errors so we could not receive answers. My exam grade for one exam changed 3 times."
121 comments	"Slides are too dense which makes it impossible to understand the key points. He does not break down the material in a simple manner."
Most neutral ratings [−0.6 to +0.6)	"She was overall fair when teaching the course. She gave extra practice problems. However did come off as very rude when asking a question or a dilemma was presented."
114 comments	"Wasn't very effective in teaching the material. He obviously knew what he was talking about"
Second most positive ratings [+0.6 to +1.8)	"Overall this instructor was very effective. However on a few rare occasions some seemingly important topics were rushed in lecture."
322 comments	"He was a good professor"
Most positive ratings [+1.8 to +3]	"He was very clear and concise and taught exactly what we needed to know"
378 comments	"She was one of the best instructors I have had at school. She is very effective in teaching the course and helping her students retain the information. She is very welcoming to coming for extra help as well if needed. She clearly knows what she is talking about as well. She was a fantastic teacher and I will not hesitate to take a class with her again."

Note. Student comments were deidentified but otherwise left unedited and, therefore, may contain typos.

more positive light). However, it is more likely that this positive skew reflects the types of comments left on course evaluations as these are real student comments taken from real evaluations.

3.2. Comparison of AI-generated ratings to human ratings

Each of our four AI models was asked to generate a sentiment score for each of the 500 comments in the testing dataset. This was done with the models in a naïve state, as well as for 10 cases of fine-tuning. We then calculated the Pearson's correlation coefficient (*r*) between each AI output and our human sentiment ratings for the testing comments. This coefficient measures the linear relationship between the AI's predicted sentiment and the human-assigned scores, where a value of 1.0 would indicate perfect agreement and 0 would indicate no relationship. This allows us to quantify the "reliability" or consistency of the AI's judgment against the human gold standard and address our research questions: can AI perform sentiment analysis of course evaluations equivalent to humans (RQ1), does fine-tuning improve performance (RQ2), and are there differences in performance across AI models (RQ3).

Without training, naïve BERT performed poorly with an initial *r* of −0.061. However, when fine-tuned, BERT's performance increased, achieving an *r* > 0.7 when given at least 250 training comments and hovering near a *r* of 0.8 after 450 training comments (see Fig. 2). These correlation coefficients are on par with what is typically considered good inter-rater reliability amongst human observers (Koo & Li, 2016; McHugh, 2012).

Even without training from our dataset, naïve RoBERTa displayed a

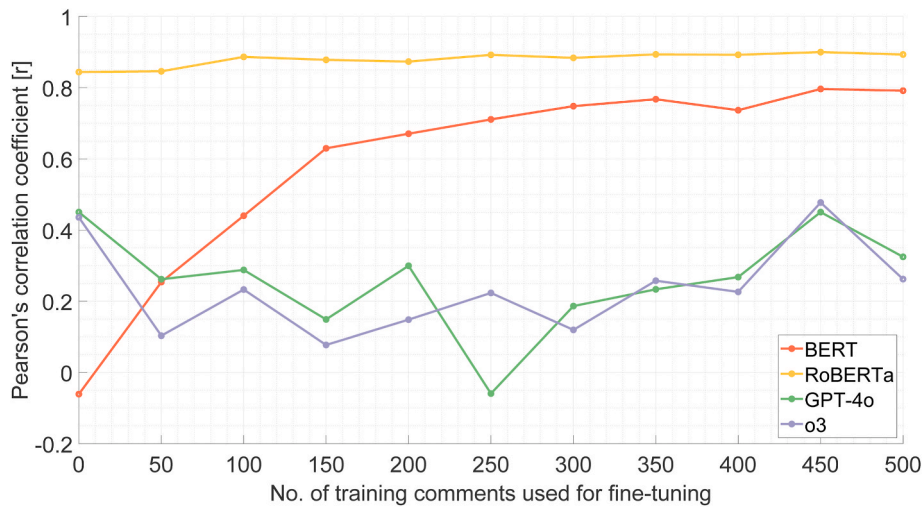


Fig. 2. Comparison of AI-generated ratings to human ratings with variable training data. Zero training comments represents the models in their naïve state without any fine-tuning. The y-axis is the Pearson's correlation coefficient between the AI sentiment scores of the testing comments and the average human ratings of the same comments.

robust similarity to human raters ($r = 0.844$). When fine-tuned, RoBERTa showed minimal increases in improvement. Fine-tuning cases of 100–500 training comments resulted in correlations (r) between RoBERTa's output and the human sentiment scores that ranged from 0.873 to 0.900. This suggests that RoBERTa had quickly reached its peak performance.

OpenAI's GPT-4o and o3 performed in similar unexpected ways. Naïve models both began with a Pearson's correlation coefficient slightly above 0.4 (0.450 and 0.436 for GPT-4o and o3, respectively). These initial correlations were better than BERT but poorer than RoBERTa and poorer than what researchers typically consider an acceptable inter-rater reliability (Koo & Li, 2016; McHugh, 2012). However, neither model improved with training. In most cases, attempts at fine-tuning seemed to be detrimental to the models' performance on sentiment analysis.

3.3. Testing the reliability of the AI models

Given the unexpected findings from GPT-4o and o3, we conducted a reliability analysis of all four AI models. We ran the naïve version of each prompt 10 times and, once again, calculated Pearson's r between each AI

output and the human ratings. Each model performed consistently across all 10 iterations and consistent with the naïve output from the previous analysis (see Fig. 3). These findings suggest that both MLMs and GLMs remain largely stable in their sentiment predictions across distinct and unique testing instances (with an especially stable trend observed for RoBERTa and GPT-4o) as long as they are given the same inputs (or in the case of the GLMs, the same prompt) each time. However, for the generative models, when the prompt was the same but the amounts of training data differed, such as in the 50 comment increments, there was a consistent lack of progress in sentiment evaluation. Thus, it is possible that the decision-making happening in the black box models from OpenAI is more capricious than one might expect.

3.4. Comment length distributions across sentiment scores and AI models

Due to the context-dependent nature of NLP, it is possible that longer comments provide additional context that improves the sentiment analysis compared to the analysis of shorter comments. To address whether or not comment length was biasing our sentiment predictions, we examined the relationship between comment length and AI-generated sentiment scores across training instances. Most student

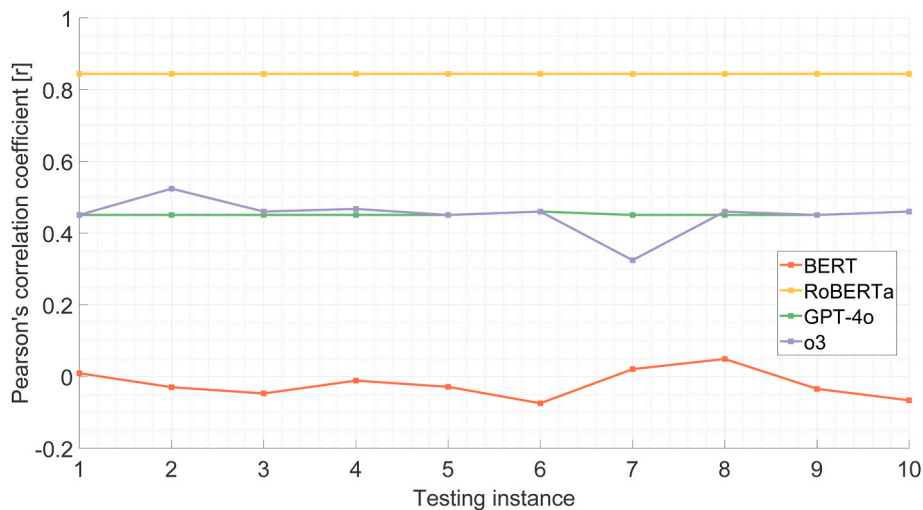


Fig. 3. A reliability analysis of the AI models. Ten separate instances of the naïve models were run on the testing set of comments. These AI-generated sentiment scores were then compared to the average human ratings. Across all ten instances, the models appeared stable as there was little change in their performance.

comments were under 50 words; only 5.6 % of the testing comments were greater than 50 words in length (range = 1–187, median = 13.5). When comparing the average sentiment score for the lowest fifth of words per comment (WPC, 1–5 words) to the highest fifth of WPC (more than 31 words), all AI models and training cases showed little change in sentiment as WPC increased (see Fig. 4), indicating that comment length was not biasing the AI-generated sentiment scores. The exception was BERT fine-tuned with 50 comments, and to a lesser extent BERT fine-tuned with 100 comments, for which sentiment increased as comment length increased. Given that our previous analyses showed that these cases of fine-tuning BERT resulted in a performance far lower than when BERT was trained with more data (see Fig. 2), it appears that the poor performance in these two instances may, in part, be due to a misguided attempt by the AI model to gauge sentiment from comment length.

3.5. Computational time for model performances

When considering possible adoption of some of the investigated LLMs in large scale usage for applications like higher education, computational time becomes an important factor to note. Fig. 5 quantifies the computational cost of AI-powered sentiment analysis in training and scoring comments, comparing the performance of BERT and RoBERTa across increasing training data sizes. Training refers to the fine-tuning process of taking a pre-trained naïve model and providing it with a small, specialized dataset (e.g., human ratings of course evaluation comments) to "tune" its parameters for our specific classification task. We did not assess the computational times for the GLMs since the sentiment performance in these models did not confidently reflect that the models were learning. As expected, training time rises steadily with the number of comments used for fine-tuning, with RoBERTa requiring more time than BERT at higher data volumes (e.g., training off 500 comments takes RoBERTa about three times longer than BERT). This scaling trend underscores the importance of model selection in large-scale applications, where training costs can become a bottleneck. However, another factor to consider is that, given naïve RoBERTa's robust performance in our previous analyses, RoBERTa may need little

to no fine-tuning. Scoring time remains relatively low and stable for both models (though BERT is still about three times faster than RoBERTa), highlighting their feasibility for deployment in real-time or batch sentiment evaluation tasks.

4. Discussion

Although there is much interest from the field of higher education in the data supplied from course evaluations, the qualitative student comments on these evaluations are often excluded from analysis and from research studies. Qualitative research is labor-intensive, which has likely limited this area of exploration. Thus, the primary goal of the current study was to investigate whether or not AI models can perform this analysis in a manner equivalent to humans (RQ1). Relatedly, we also examined whether or not fine-tuning the models improved their performance (RQ2) and whether or not there were differences in performance between models (RQ3).

In response to RQ1, our comparison of four different AI models revealed that some models are able to analyze the student comments written on course evaluations at a level that is on par with human raters. In the most promising case, RoBERTa performed robustly even without any specific fine-tuning. Whether or not fine-tuning improved performance varied by AI model (RQ2). Training improved BERT's performance in a monotonic fashion with larger training sets leading to greater improvements, though there appeared to be some leveling off after 450 comments. Fine-tuning led to minimal improvements in RoBERTa, most likely due to RoBERTa quickly reaching peak performance. Fine-tuning had little to no effect on GPT-4o and o3, suggesting that these models are not able to learn from feedback, at least as when given instructions via their chatbot windows. In answering RQ1 and RQ2, we have also addressed RQ3. There are clearly differences amongst the models as not all models could conduct a satisfactory sentiment analysis and some models improved with training while others did not.

Taken together, there are two important findings from our study. The first is that LLMs can be a solution to the time and labor required to analyze large amounts of qualitative data and that this opens up a nearly

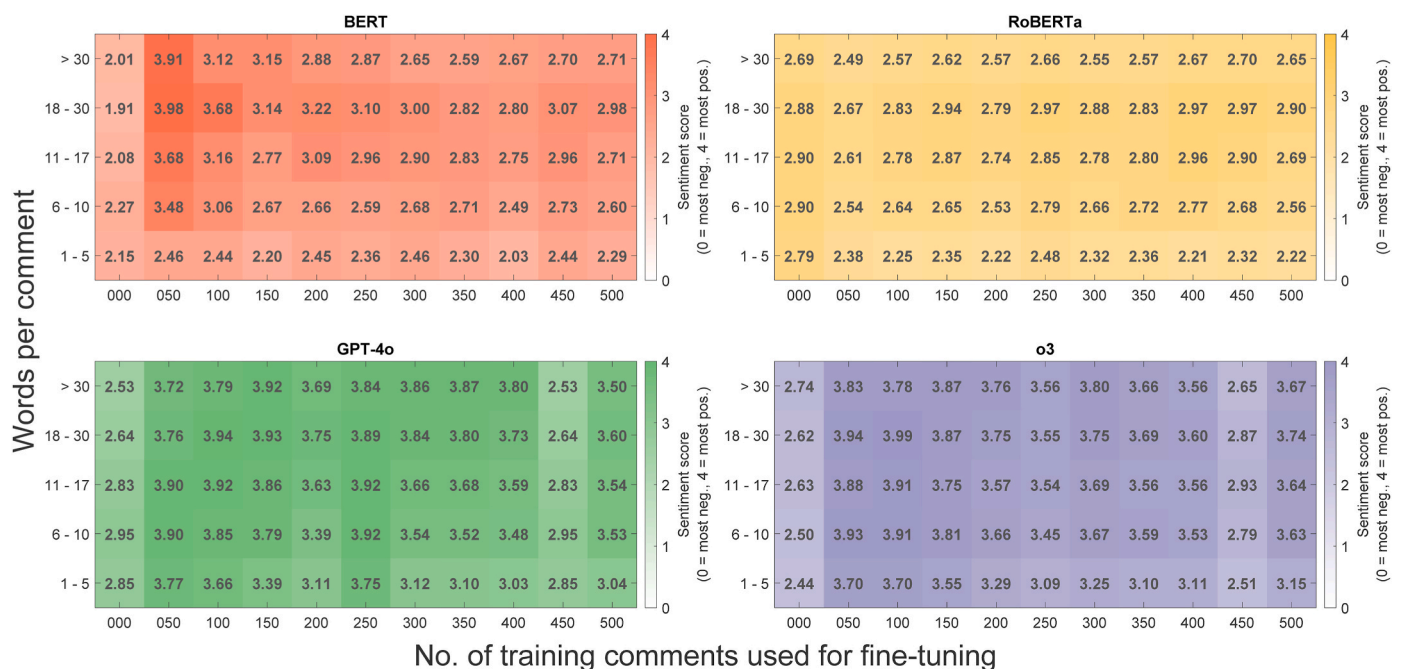


Fig. 4. Distributions of AI-generated sentiment scores as a function of comment length. As comment length increased (from 1 to 5 words to more than 30 words), there was little change in the average AI-generated sentiment score, indicating that comment length was not biasing the AI output. An exception was BERT fine-tuned with 50 comments, and to a lesser extent, BERT fine-tuned with 100 comments. However, previous analyses revealed that these specific cases performed poorly on sentiment analysis.

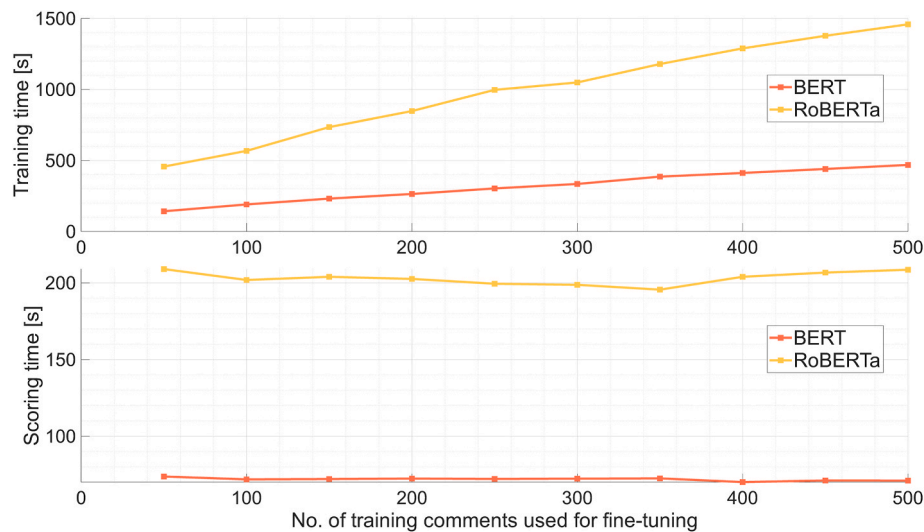


Fig. 5. Computational time (in seconds) for sentiment analysis (training + scoring) across increasing training sizes for BERT and RoBERTa. Training time (top panel) increases with training dataset size. Scoring times (bottom panel) were considerably faster than training times. Computational times for GPT-4o and o3 were not included since the sentiment performance in these models did not confidently reflect that the models were learning.

untouched mine of data as well as the potential to analyze very large sets of data (e.g., across an entire institution, across multiple institutions, or across several years of an institution's course evaluations). However, since not all of our AI models were able to provide a reliable sentiment analysis, the second major finding is an important reminder that individuals and institutions should carefully consider what AI model to use for a given task as this has implications on the reliability and validity of the results. We expand on these points below.

4.1. The use of AI increases possible avenues of course evaluations analysis

There has been much research on factors (e.g., instructor, course, or student variables) that are associated with various ratings on course evaluations. However, to the extent of our knowledge, all previous research has been conducted on the quantitative ratings on evaluations. Although favored for their ease of statistical analysis, Likert-scale data has inherent limitations. One of which is an artificiality or lack of representativeness of the participants' true response due to participants having to constrain themselves to a limited number (typically, five) of generic responses. Participants are unable to respond in their own words but, instead, must choose amongst preselected options (e.g., “agree” or “disagree”). Qualitative responses, which are typically more detailed and nuanced, may be more informative than quantitative ratings as the additional detail may differentiate whether a student's evaluation is a valid assessment of instructor “effectiveness” as opposed to merely an indication of the student's “satisfaction” with their course grade.

Another possible limitation of the rating scale data from course evaluations is a ceiling effect in which the majority of instructors are rated very highly. There may be many reasons for this ceiling effect. For example, it may be an artifact of low response rates (Fincham, 2008). However, regardless of the cause, which is beyond the focus of this study, a ceiling effect makes it difficult to draw generalizable conclusions from course evaluation data. Being able to analyze student comments instead of relying on the Likert-scale ratings could potentially address these issues. Although sentiment analysis converts the comments into an integer, these sentiment ratings may be a more accurate representation of the participants' responses, and they may reduce the ceiling effect which has benefits for course evaluation research. It is still an open question as to whether or not the qualitative student comments are more informative than the quantitative Likert-scale ratings. However, to answer that question, we first need a way to analyze the

qualitative comments, and the current study demonstrates a proof-of-concept that AI can perform this function.

The efficiency of AI also allows for the possibility of examining large course evaluation datasets, such as those across an entire institution, across multiple institutions, or across multiple years of course evaluation data. These large datasets would allow researchers to better investigate patterns of course evaluation data, for example, comparing various instructor, student, and course factors such as teaching experience and a greater range of class sizes. Relatedly, instructor, student, and course factors all change with time and are influenced by the prevailing culture of the day. For example, there are a greater number of female and minority faculty than there were in the 1920s or 1970s. Likewise, there are more female and minority students. The COVID-19 pandemic and aftermath illustrated how quickly many factors related to higher education may shift (e.g., technology, class format, student populations, college preparedness, etc.). AI provides a way to investigate the qualitative data of large sets of course evaluations over time.

4.2. The importance of carefully selecting an AI model for a given task

Regardless of whether one views the prospects of AI with excitement or reservation, it is being used in many different fields including higher education and research (e.g., Crompton & Burke, 2023). However, it is important to carefully consider which AI model is being used. The average individual is likely most familiar with ChatGPT (currently, OpenAI's GPT-4o) and may gravitate towards that AI model. However, in our analysis, neither GPT-4o and o3 were capable of performing a sentiment analysis on par with humans. In particular, this study identified that they were unable to learn from feedback or training. Attempting to answer RQ1 and RQ2 with only ChatGPT would have led to very different conclusions. Thus, it is important to be aware that not all AI models are the same and that they cannot be used interchangeably with one another.

Our results indicate that both BERT and RoBERTa are satisfactory models to use for sentiment analysis. RoBERTa has much higher alignment with human performance from the start and needs minimal training, though one limitation may be that it may not have the flexibility to be easily tunable with an additional dataset (since it was already so fine-tuned to begin with). Also, as one trains it with more comments, the computational expense starts climbing. If that additional flexibility is needed for some applications, the option to train BERT on an individual's or institution's own cultivated dataset may be just as good as a

choice as RoBERTa, with the benefit of a lower computational cost.

4.3. Limitations and other considerations

In order to maximize external validity, we chose to use real student comments written on course evaluations. For this reason, the comments we used may be biased in some manner. For example, we did see a positive skew in the comments. However, we believe this positive skew is descriptive of a real phenomenon and more appropriate to study than a dataset without any skew (e.g., an artificially created set of comments that were equally positive and negative). We took multiple steps to minimize potential bias and to create as representative a set of comments as possible, for example, selecting comments from a wide range of common disciplines. We intentionally avoided a participant pool of solely psychology students (as is typically used in many studies involving human participants), and we monetarily compensated participants to increase motivation (as opposed to having participants complete the survey simply to fulfill a class requirement). Thus, we believe we have preserved the validity of our curated set of comments and human sentiment ratings.

All the course evaluation comments used in this study and human raters of those comments came from one institution. This, of course, can raise the question of generalizability since it is possible that the course evaluations used in this study are not representative of all course evaluations. We believe the institution these comments came from is representative of higher education as it is a large state university in the United States. However, we are unable to directly address this question since we do not have access to other institutions' data. What course evaluation data looks like across institutions is an interesting and relevant question, and we encourage further research on this topic. Perhaps our discovery of AI as a useful tool to analyze student comments will aid this research.

OpenAI's GPT-4o and o3 are not open source. Therefore, it is difficult to determine how these models are responding to our task instructions. Fine-tuning these models may require more specific prompt engineering and trial-and-error. However, we thought it important to include these models to provide a comparison of MLMs versus GLMs. Additionally, ChatGPT (GPT-4o) is currently among the most well-known and widely used LLMs amongst the general public, and thus, assessing its performance on a task like sentiment analysis seemed integral to the themes of our research study. A related consideration to take into account is that the underlying base of GPT-4o and o3 may change over time as OpenAI makes updates and increases the models' pretraining data. This has implications for our GLM results as OpenAI may adjust the models in ways that improve their sentiment analysis capability, but also implications on AI model selection, generally speaking. LLM users may not realize that they are using slightly different versions of GPT-4o or o3 over time. Thus, any changes in the results may actually be a reflection of changes in the instrument which can threaten the validity of the results. We recognize these limitations and believe they are important for other AI users to be aware of as well.

Finally, while this study utilizes Pearson's correlation to establish a foundational alignment benchmark, we acknowledge that more advanced statistical frameworks could offer further nuance. Future research might employ Generalizability Theory (Cronbach et al., 1972) to decompose sources of variance between models, or Bland-Altman analysis (Bland & Altman, 1986) to specifically quantify systematic bias across the sentiment scale. However, for the present proof-of-concept, the product-moment correlation remains the most transparent and validated metric for establishing initial linear alignment between human and AI raters.

4.4. Domain expansion of the theoretical framework

Although this study focused on the alignment of sentiment judgments between AI models and human raters in the context of course evaluations, the proposed framework readily generalizes to scientific

settings where researchers face high-volume open-ended text that is costly to code manually (e.g., surveys, interviews, focus groups, feedback forms, and incident narratives). In these settings, an LLM can be positioned as a scalable first-pass coder that applies an explicit, pre-defined label space (e.g., a set of ratings/categories), with the LLM's performance validated against human judgments before being used for inference. This framing aligns with emerging methodological work that treats LLMs as supports for structured deductive coding rather than replacements for qualitative interpretation (Tai et al., 2024).

Another domain expansion of this framework is into applied settings where qualitative text already exists but is difficult to analyze at scale. Examples include: (1) safety and incident reporting, where narratives can be screened or coded into event types and/or severity ratings; (2) maintenance and reliability, where work orders can be classified into categories or urgency levels to support triage and staffing; and (3) technical corpora, where LLMs can be domain-adapted via fine-tuning and can enable large-scale synthesis on highly technical text beyond what manual review permits. Across these domains, the transfer of the framework is straightforward: replace "sentiment" with a domain-relevant construct (e.g., severity, confidence, risk, or impact), define it with a clear rubric/codebook, validate on a modest human-coded benchmark, and then deploy for screening/triage and workload reduction. Domain-adapted pretraining is a practical extension of this framework when specialized jargon drives errors, as previously demonstrated in the literature in some scientific and aviation-specific BERT variants (Andrade & Walsh, 2023; Beltagy et al., 2019).

However, this framework expansion must also be grounded in the distinction between algorithmic sentiment detection and educational interpretation. While the product-moment correlation (Pearson, 1896) validates the analytical capability of LLMs to replicate human scoring, the theoretical application requires a "human-in-the-loop". From a pedagogical standpoint, we must consider the theory of "pedagogical tact" (van Manen, 2016). Education is fundamentally a relational endeavor; therefore, while AI-driven analytics can identify what a student is feeling at scale, the human educator remains indispensable for understanding why that sentiment matters within the specific social context of the learning environment. This theoretical expansion positions AI as a supportive tool for real-time pedagogical analytics rather than a total replacement for human evaluative judgment.

5. Conclusion

In summary, we demonstrate that AI can conduct a sentiment analysis of the qualitative student comments written on course evaluations in a manner equivalent to human raters. This has implications for future course evaluation research as this qualitative data has often been ignored and the efficiency of AI allows for the possibility of analyzing large sets of course evaluation data, a resource already available at many institutions of higher education. However, our study also compared multiple AI models, and we caution that not all AIs perform the same. In particular, ChatGPT (currently, OpenAI's GPT-4o) was not able to perform a reliable sentiment analysis. It is important to be aware of ChatGPT's limitations as its familiarity may make it the default choice for many. Individuals and institutions should carefully consider which AI model is used for a given task as this has implications on the validity of their findings. Ultimately, the innovation of this approach lies in its ability to augment, rather than replace, human expertise. By elucidating when there may be areas of "selective alignment", we provide a framework where AI handles the massive scale of initial quantification of qualitative data, allowing human educational experts to focus their attention on the nuanced outliers where AI and human judgment diverge – the very areas where the most critical pedagogical insights often reside.

CRediT authorship contribution statement

Joyce W. Lacy: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Chi Nnoka:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zachary Jock:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Cathleen Morreale:** Resources, Data curation, Conceptualization.

Ethics

This study was approved by our institutional ethics committee with IRB ID #00008366. Informed consent was obtained from all participants, and their privacy rights were strictly observed.

Data availability

Data will be made available on request.

Funding

Data collection from human participants was funded by a seed grant from the University at Buffalo, Office of Curriculum, Assessment and Teaching Transformation received by JWL.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Andrade, S. R., & Walsh, H. S. (2023). SafeAeroBERT: Towards a safety-informed aerospace-specific language model. In *AIAA AVIATION 2023 Forum* (p. 3437).
- Aragon, O. R., Pietri, E. S., & Powell, B. A. (2023). Gender bias in teaching evaluations: The causal role of department gender composition. *Proceedings of the National Academy of Sciences*, 124(4), Article e2118466120.
- Backer, E. (2012). Burnt at the student evaluation stake – The penalty for failing students. *E-Journal of Business Education & Scholarship of Teaching*, 6(1), 1–13.
- Bahdanau, D., Cho, K., & Bengio, Y. (2016). Neural machine translation by jointly learning to align and translate. *arXiv*. <https://arxiv.org/abs/1409.0473>.
- Beltagy, I., Lo, K., & Cohan, A. (2019). SciBERT: A pretrained language model for scientific text. *arXiv preprint arXiv:1903.10676*.
- Benton, S. L., & Li, D. (2015). Validity and reliability of IDEA teaching essentials (IDEA Research Report No. 8). *The IDEA Center*.
- Benton, S. L., Li, D., Brown, R., Guo, M., & Sullivan, P. (2015). *Revising the IDEA student ratings of instruction system 2002-2011 data (IDEA Technical Report No.1 8)*. The IDEA Center.
- Beran, T. N., & Rokosh, J. L. (2009). Instructors' perspectives on the utility of student ratings of instruction. *Instructional Science*, 37, 171–184. <https://doi.org/10.1007/s11251-007-9045-2>.
- Bland, J. M., & Altman, D. (1986). Statistical methods for assessing agreement between two methods of clinical measurement. *The Lancet*, 327(8476), 307–310.
- Bravais, A. (1846). Analyse mathématique sur les probabilités des erreurs de situation d'un point. *Mémoires présentés par divers savants à l'Académie royale des sciences de l'Institut de France*, 9, 255–332.
- Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D., Wu, J., Winter, C., ... Amodi, D. (2020). Language models are few-shot learners. In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, & H. Lin (Eds.), *Advances in neural information processing systems* (Vol. 33, pp. 1877–1901). Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2020/file/1457c0d6bfc94967418bf8ac142f64a-Paper.pdf.
- Centra, J. A. (2009). *Differences in responses to the student instructional report: Is it bias?* Educational Testing Service.
- Cho, D., Baek, W., & Cho, J. (2015). Why do good performing students highly rate their instructors? Evidence from a natural experiment. *Economics of Education Review*, 49, 172–179. <https://doi.org/10.1016/j.econedurev.2015.10.001>
- Crompton, H., & Burke, D. (2023). Artificial intelligence in higher education: The state of the field. *International Journal of Educational Technology in Higher Education*, 20. <https://doi.org/10.1186/s41239-023-00392-8>. Article 22.
- Cronbach, L. J., Gleser, G. C., Nanda, H., & Rajaratnam, N. (1972). *The dependability of behavioral measurements: Theory of generalizability for scores and profiles*. John Wiley & Sons.
- Devlin, J., Chang, M., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv*. <https://doi.org/10.48550/arxiv.1810.04805>
- Fincham, J. E. (2008). Response rates and responsiveness for surveys, standards, and the journal. *American Journal of Pharmaceutical Education*, 72(2), 1–3. <https://doi.org/10.5688/aj720243>
- Flick, U. (2009). *An introduction to qualitative research* (4th ed.). Sage.
- Galton, F. (1888). Co-relations and their measurement, chiefly from anthropometric data. *Proceedings of the Royal Society of London*, 45, 135–145.
- Gokaslan, A., & Cohen, V. (2019). OpenWebText Corpus. <http://skylion007.github.io/OpenWebTextCorpus>.
- Hamilton, L., Elliott, D., Quick, A., Smith, S., & Choplin, V. (2023). Exploring the use of AI in qualitative analysis: A comparative study of guaranteed income data. *International Journal of Qualitative Methods*, 22, 1–13.
- Hoyt, D. P., & Lee, E.-J. (2003). *Understanding the IDEA system's extraneous variables (IDEA Research Report No. 6)*. The IDEA Center.
- Hurst, A., Lerer, A., Goucher, A. P., Perelman, A., Ramesh, A., Clark, A., Ostrow, A., Welihinda, A., Hayes, A., Radford, A., et al. (2024). GPT-4o system card. *arXiv*. <https://arxiv.org/abs/2410.21276>.
- Jurafsky, D., & Martin, J. H. (2008). *Speech and language processing: An introduction to natural language processing, computational linguistics, and speech recognition* (2nd ed.). Prentice Hall.
- Kamath, U., Keenan, K., Somers, G., & Sorenson, S. (2024). *Large language models: A deep dive*. Springer.
- Kebede, D., & Tesfai, N. (2023). *AI-powered text analysis tool for sentiment analysis* [Master's thesis. Mälardalen University]. DiVA Portal. <https://urn.kb.se/resolve?urn=urn:nbn:se:mdh:diva-63059>.
- Kerner, S. M. (2025). OpenAI o3 and o4 explained: Everything you need to know. *TechTarget*. <https://www.techtarget.com/whatis/feature/OpenAI-o3-explained-Everything-you-need-to-know>.
- King, N., & Brooks, J. (2018). Thematic analysis in organisational research. In C. Cassell, A. Cunliffe, & G. Grandy (Eds.), *The SAGE handbook of qualitative business and management research methods: Methods and challenges* (Vol. 2, pp. 219–236). Sage.
- Koo, T. K., & Li, M. Y. (2016). A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *Journal of Chiropractic Medicine*, 15, 115–163.
- Li, D., & Benton, S. L. (2017). *The effects of instructor gender and discipline group on student ratings of instruction (IDEA Research Report No. 10)*. The IDEA Center.
- Likert, R. (1932). A technique for the measurement of attitudes. *Archives of Psychology*, 22(140), 1–55.
- Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). RoBERTa: A robustly optimized BERT pretraining approach. *arXiv*. <https://doi.org/10.48550/arxiv.1907.11692>
- Liu, Z., Qiao, A., Neiswanger, W., Wang, H., Tan, B., Tao, T., Li, J., Wang, Y., Sun, S., Pangarkar, O., Fan, R., et al. (2023). LLM360: Towards fully transparent open-source LLMs. *arXiv preprint arXiv:2312.06550*.
- McHugh, M. L. (2012). Interrater reliability: The kappa statistics. *Biochemia Medica*, 22(3), 276–282.
- Mengel, F., Sauermann, J., & Zolitz, U. (2019). Gender bias in teaching evaluation. *Journal of the European Economic Association*, 17(2), 535–566. <https://doi.org/10.1093/jea/evx057>
- Miller, J. E., & Seldin, P. (2014). *Changing Practices in Faculty Evaluation: Can better evaluation make a difference?* American Association of University Professors. <http://www.aaup.org/article/changing-practices-faculty-evaluation#.VuIyE0UWpo>.
- Nagel, S. (2016). CC-News: Common crawl news dataset. <https://commoncrawl.org/2016/10/news-dataset-available/>.
- Onan, A. (2020). Sentiment analysis on massive open online course evaluations: A text mining and deep learning approach. *Computer Applications in Engineering Education*, 29(3), 572–589.
- OpenAI. (2025). Introducing o3 and o4-mini. <https://openai.com/index/introducing-o3-and-o4-mini/>.
- Patten, M. L. (1998). *Questionnaire research: A practical guide*. Los Angeles: Pryczak.
- Pearson, K. (1896). Mathematical contributions to the theory of evolution. III. Regression, heredity, and panmixia. *Philosophical Transactions of the Royal Society of London - Series A: Containing Papers of a Mathematical or Physical Character*, 187, 253–318.
- Pfeifer, M. A., & Dolan, E. L. (2023). Venturing into qualitative research: A practical guide to getting started. *Scholarship and Practice of Undergraduate Research*, 7(1), 10–20.
- Seetharaman, D. (2024). OpenAI launches voice assistant inspired by Hollywood vision of AI. *The Wall Street Journal*. <https://www.wsj.com/tech/ai/openai-voice-assistant-chatgpt-app-376>.
- Seldin, P. (1998). How colleges evaluate teaching: 1988 vs. 1998. *American Association of Higher Education Bulletin*, 50, 3–7.
- Shannon, C. E. (1948). A mathematical theory of communication. *The Bell System Technical Journal*, 27, 379–423. <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>
- Silverman, D. (2000). *Doing qualitative research*. Sage.
- Smith, B. P. (2007). Student ratings of teacher effectiveness: An analysis of end-of-course faculty evaluation. *College Student Journal*, 41, 788–800.

- Steyn, C., Davies, C., & Aerni, S. (2019). Eliciting student feedback for course development: The application of a qualitative course evaluation tool among business research students. *Assessment & Evaluation in Higher Education*, 44(1), 11–24. <https://doi.org/10.1080/02602938.2018.1466266>
- Stroebe, W. (2020). Student evaluations of teaching encourages poor teaching and contributes to grade inflation: A theoretical and empirical analysis. *Basic and Applied Social Psychology*, 42(4), 276–294.
- Tai, R. H., Bentley, L. R., Xia, X., Sitt, J. M., Fankhauser, S. C., Chicas-Mosier, A. M., & Monteith, B. G. (2024). An examination of the use of large language models to aid analysis of textual data. *International Journal of Qualitative Methods*, 23, Article 16094069241231168.
- Tai, K. S., Socher, R., & Manning, C. D. (2015). Improved semantic representations from Tree-Structured Long Short-Term memory networks. *arXiv*. <https://doi.org/10.48550/arxiv.1503.00075>
- Tornberg, P. (2023). ChatGPT-4 outperforms experts and crowd workers in annotating political Twitter messages with zero-shot learning. *arXiv*. <https://arxiv.org/abs/2304.06588>.
- Trinh, T. H., & Le, Q. V. (2019). A simple method for commonsense reasoning. *arXiv*. <https://arxiv.org/abs/1806.02847>.
- van Manen, M. (2016). *The tact of teaching: The meaning of pedagogical thoughtfulness*. Routledge.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, & R. Garnett (Eds.), *Advances in neural information processing systems* (Vol. 30). Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf.
- Wang, G., & Williamson, A. (2022). Course evaluation scores: Valid measures for teaching effectiveness or rewards for lenient grading? *Teaching in Higher Education*, 27(3), 297–318.
- Zhu, Y., Kiros, R., Zemel, R., Salakhutdinov, R., Urtasun, R., Torralba, A., & Fidler, S. (2015). Towards story-like visual explanations by watching movies and reading books. In *Proceedings of the IEEE international conference on computer vision (ICCV)* (pp. 3355–3363). IEEE. <https://doi.org/10.1109/ICCV.2015.384>.